Classwork: Pearson Correlation, Spearman Correlation, and Simple Linear Regression

Statistics ($n = 20$)

Instructions. Complete every part *by hand*, showing all formulas and intermediate work. Then reproduce all results *in R* and attach your script and output. Round final answers to four decimal places.

The data

A researcher records, for 20 students, the number of hours studied (X) and the exam score out of 100 (Y).

i	X_i	Y_i	i	X_i	Y_i
1	3	45	11	19	75
2	5	52	12	20	88
3	6	48	13	22	85
4	8	61	14	23	92
5	10	58	15	25	90
6	11	67	16	26	95
7	13	70	17	28	89
8	14	65	18	30	98
9	16	78	19	32	96
10	17	82	20	35	99

1 Preliminary sums

Compute and report each of the following quantities. Show the arithmetic.

(a) $\sum X, \quad \sum Y, \quad \bar{X}, \quad \bar{Y}$

(b) $\sum X^2, \quad \sum Y^2, \quad \sum XY$

(c) The corrected sums of squares and cross-products:

$$S_{xx} = \sum X^2 - \frac{(\sum X)^2}{n}, \qquad S_{yy} = \sum Y^2 - \frac{(\sum Y)^2}{n}, \qquad S_{xy} = \sum XY - \frac{(\sum X)(\sum Y)}{n}.$$

2 Pearson correlation coefficient

(a) Using your values above, compute

$$r = \frac{S_{xy}}{\sqrt{S_{xx}\, S_{yy}}}.$$

(b) Compute r^2 and state, in one sentence, what it means in context.

(c) Describe the direction and strength of the linear relationship.

3 Spearman rank correlation coefficient

Rank X and Y separately from smallest (1) to largest (20). Fill in the table, then compute

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2-1)}, \qquad d_i = \operatorname{rank}(X_i) - \operatorname{rank}(Y_i).$$

i	X_i	Y_i	$\operatorname{rank} X$	$\operatorname{rank} Y$	d_i	d_i^2
1	3	45				
2	5	52				
3	6	48				
4	8	61				
5	10	58				
6	11	67				
7	13	70				
8	14	65				
9	16	78				
10	17	82				
11	19	75				
12	20	88				
13	22	85				
14	23	92				
15	25	90				
16	26	95				
17	28	89				
18	30	98				
19	32	96				
20	35	99				
					$\sum d_i^2 =$	

(a) Report $\sum d_i^2$ and compute ρ.

(b) Is ρ equal to 1.00? Explain what a value below 1.00 tells you about the ordering of the data.

(c) Compare r and ρ. Which is larger here, and what does that suggest about linearity versus monotonicity?

4 Simple linear regression of Y on X

(a) Compute the least-squares slope and intercept:

$$b_1 = \frac{S_{xy}}{S_{xx}}, \qquad b_0 = \bar{Y} - b_1 \bar{X},$$

and write out the fitted equation $\hat{Y} = b_0 + b_1 X$.

(b) Interpret the slope b_1 in context (one sentence). Interpret the intercept b_0, and note whether it is an extrapolation.

(c) Compute the partition of variability: $\text{SST} = S_{yy}$, SSR, and SSE. Verify that $R^2 = \text{SSR}/\text{SST}$ equals r^2 from Part 2.

(d) Compute the residual variance $s^2 = \text{SSE}/(n-2)$ and s.

(e) Test $H_0 : \beta_1 = 0$ against $H_1 : \beta_1 \neq 0$ at $\alpha = 0.05$. Compute

$$\text{SE}(b_1) = \frac{s}{\sqrt{S_{xx}}}, \qquad t = \frac{b_1}{\text{SE}(b_1)},$$

report the degrees of freedom, and state your conclusion.

(f) Predict the exam score for a student who studies $X = 21$ hours.

5 Do it all in R or STATA

Enter the data and reproduce every result above in R. At minimum, your script should:

(a) Create the vectors `X` and `Y`.

(b) Compute S_{xx}, S_{yy}, S_{xy} directly from the sums.

(c) Compute the Pearson correlation two ways: from the formula and with `cor(X, Y, method = "pearson")`. Confirm they match.

(d) Compute the Spearman correlation two ways: from the $\sum d_i^2$ formula (using `rank()`) and with `cor(X, Y, method = "spearman")`. Confirm they match.

(e) Fit the regression with `lm(Y ~ X)` and report `summary()`. Identify b_0, b_1, s, R^2, the t-statistic for the slope, and its p-value in the output.

(f) Use `predict()` to obtain the fitted value at $X = 21$ and confirm it matches your hand calculation.

(g) Produce a scatterplot of Y against X with the fitted regression line added.

Attach your R script and its full console output. Your hand results and R results should agree.